LU Talk for November, 2025

Language Understanding λ(μ) Laboratory

FROM PIXELS TO PROGRESS: AI AND COMPUTER VISION FOR A SMARTER, SAFER WORLD

Artificial Intelligence and Computer Vision have rapidly evolved from theoretical concepts into powerful technologies shaping public safety, healthcare, and daily life. This work explores how deep learning, particularly spatiotemporal modelling, enables machines to perceive, interpret, and act upon complex visual environments with unprecedented accuracy. Research on anomaly detection in surveillance videos using Deep Spatiotemporal Translation Networks (DSTN) and their enhanced variant (DR-STN) significantly improves the reliability of abnormal event detection across challenging real-world datasets. These models address long-standing issues such as noise, motion ambiguity, and misdetection, offering practical applications for smart cities, public security systems, and large-scale video analytics. The second case study showcases our work on AI-driven cataract detection, demonstrating how lightweight convolutional neural networks can support early diagnosis and expand access to ophthalmic care. Together, these examples illustrate the transformative potential of AI and vision technologies when applied ethically and purposefully. The work concludes with a forward-looking perspective on the role of human-centred, trustworthy AI in building intelligent systems that enhance safety, healthcare accessibility, and quality of life.

Asst. Prof. Dr. Thittaporn Ganokratanaa,

KMUTT, Thailand

Dr. Thittaporn Ganokratanaa is an Assistant Professor in the Applied Computer Science Programme at King Mongkut's University of Technology Thonburi (KMUTT). She specializes in artificial intelligence, computer vision, image processing, IoT, multimedia signal processing, and human-computer interaction. She received her Ph.D. in Electrical Engineering (Multimedia and Signal Processing) from Chulalongkorn University, with joint research collaboration at the University of Trento, Italy. She is a Senior Fellow of the UK Professional Standards Framework and Secretary of the IEEE Thailand Section.

She has published over 60 papers in international journals, conference proceedings, books, and book chapters, and has led award-winning projects such as AI-driven prosthetic arms, IoT-enabled healthcare systems, digital herbal platforms, and anomaly detection technologies. Her works have been recognized with multiple Gold and Silver medals at the International Exhibition of Inventions Geneva and the Japan Design, Idea, and Invention Expo (JDIE 2024), as well as national innovation awards from the NRCT.

30 Nov, 2025 (Sunday)
1:00 PM - 2:00 PM (Thailand Time)

Dr. Ye Kyaw Thu
Chairperson

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